HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY, VNU-HCM
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



Deep Learning and Its Applications (CO3133)

Semester 261

Assignment 2 Report

Deep Learning on Large-Scale Data and Specialized Tasks

Semantic Segmentation of Underwater Imagery:
A U-Net Baseline and Pretrained DeepLabV3 on SUIM

Group Placeholder

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Abstract

To complete: Write this last, in one paragraph of about 150 words. State the task (eight-class semantic segmentation on SUIM), the protocol (fixed split, seed, checkpoint rule, official test set), the U-Net and DeepLabV3 test mIoU and Dice, the gap between them, the outcome of the cross-entropy versus cross-entropy plus Dice experiment, the main error group, the compute trade-off, and the main limitation. Use only test-set numbers here.

Table of Contents

Abstract **i**

1 Problem and data description **1**

1.1 Task and research question 1
1.2 Dataset and study scope 1
1.3 Dataset proposal and approval 1
1.4 Exploratory data analysis 1
1.5 Splits and leakage prevention 1
1.6 Preprocessing and augmentation 3

2 Methodology **4**

2.1 Training and evaluation pipeline 4
2.2 Model architectures 4
2.3 Fine-tuning procedure 4
2.4 Loss and evaluation metrics 4
2.5 Training configuration 5
2.6 Experimental design 5

3 Implementation results **6**

3.1 Training behavior 6
3.2 Quantitative results 6
3.3 Compute cost 7
3.4 Qualitative results 7
3.5 Controlled experiment: loss function 7
3.6 Error analysis 7
3.7 Comparison and discussion 7
3.8 Limitations and conclusion 9

AI usage disclosure **10**

References **10**

1 Problem and data description

1.1 Task and research question

To complete: Give the real-world motivation for underwater scene segmentation (for example, robot navigation and marine survey). Define the input as an RGB image $x \in \mathbb{R}^{3\times H\times W}$, the output as a mask $\hat{y} \in \{0,\dots,7\}^{H\times W}$, and the prediction unit as one pixel. Write $\hat{y}_{uv} = \arg\max_k f_\theta(x)_{k,uv}$. State the research question and the main difficulties: colour cast, turbidity, small objects, and class imbalance.

1.2 Dataset and study scope

To complete: Describe SUIM [3]: source URL, version, licence status (MIT software licence; no separate dataset licence stated), collection and labelling process, the eight classes with their RGB codes, number of image and mask pairs, file formats, and storage size. Note the source overlap with EUVP, USR-248 and UFO-120 as a bias and limitation.

1.3 Dataset proposal and approval

To complete: Summarize the Dataset Proposal submitted at Milestone 1 (reports/a2_proposal.md). Record the approval status (Approved, Approved with conditions, or Rejected), the approval date, and any conditions. State whether the segmentation thresholds (at least three foreground classes and about 1,000 masked images) are met.

1.4 Exploratory data analysis

To complete: Cover the segmentation EDA items in Handbook Section 22: pixel distribution per class, region size, and mask quality. Use results/a2/eda/suim/summary.json. Report pixel share, number of images containing each class, and median region area. Explain what each figure shows and why it matters for the models.

Table 1: Per-class statistics on the usable train and validation pairs. Median area is the class fraction of an image, over images that contain the class.

Code	Class	Images	Pixel share (%)	Median area (%)
BW	Background (waterbody)	*Pending*	*Pending*	*Pending*
HD	Human divers	*Pending*	*Pending*	*Pending*
PF	Aquatic plants and sea-grass	*Pending*	*Pending*	*Pending*
WR	Wrecks and ruins	*Pending*	*Pending*	*Pending*
RO	Robots (AUVs, ROVs, instruments)	*Pending*	*Pending*	*Pending*
RI	Reefs and invertebrates	*Pending*	*Pending*	*Pending*
FV	Fish and vertebrates	*Pending*	*Pending*	*Pending*
SR	Sea-floor and rocks	*Pending*	*Pending*	*Pending*

To complete: Describe mask quality: size mismatches between image and mask, pixels whose colour is not one of the eight class codes, and how many pairs were excluded or repaired. Add a figure of example masks if it helps.

1.5 Splits and leakage prevention

To complete: State the split rule: the official test set is held out, and the remaining usable pairs are split into train and validation with seed 42. Give the split unit, whether sampling is stratified, and the saved

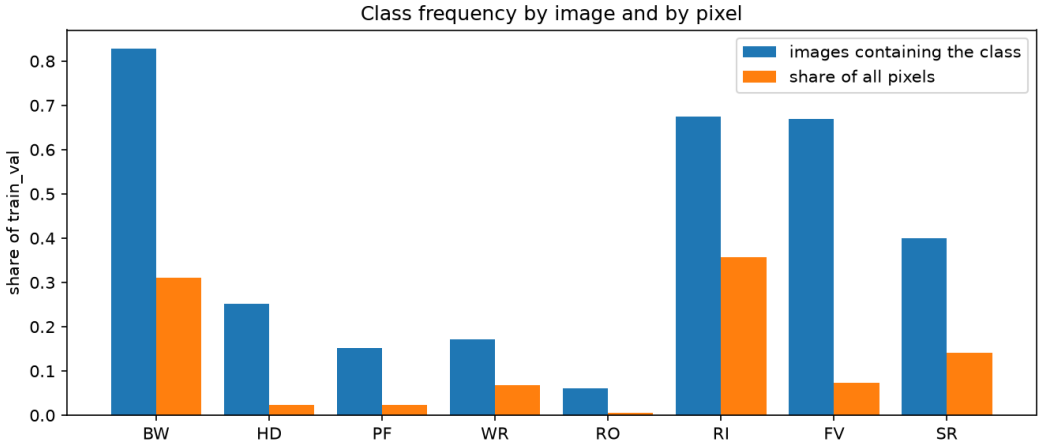


Figure 1: Pixel share and image count for each SUIM class, computed on the usable train and validation pairs.

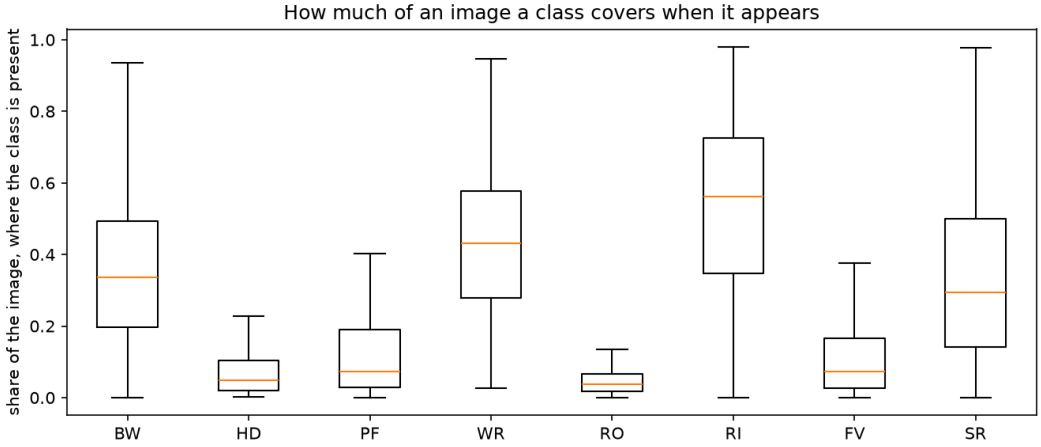


Figure 2: Distribution of region area per class, as a fraction of the image, for images that contain the class.

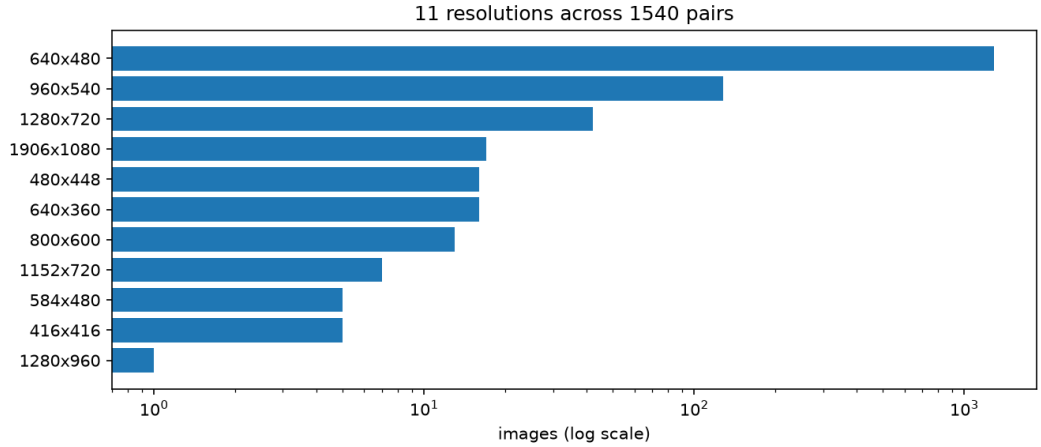


Figure 3: Distribution of native image resolutions in the train and validation pool.

split files (results/a2/splits/suim/). Explain how leakage is prevented and what has not been checked, such as near-duplicate frames from the same video or source dataset.

Table 2: SUIM splits. The official test set is held out; train and validation come from the official training pool with seed 42.

Split	Images	Source	Use
Train	*Pending*	Official train pool	Optimization
Validation	*Pending*	Official train pool	Checkpoint selection
Test	*Pending*	Official test set	Final evaluation only

1.6 Preprocessing and augmentation

To complete: Describe the conversion from RGB mask to class index, handling of off-code pixels and size mismatches, resizing to 240×320 (image with bilinear and mask with nearest-neighbour interpolation), normalization constants, and training augmentation (horizontal flip). Give the input tensor shapes and describe one example batch after preprocessing.

2 Methodology

2.1 Training and evaluation pipeline

To complete: Add a diagram of the form raw data, preprocessing, data loader, model, loss, optimization, prediction, post-processing, evaluation (Handbook Section 3.2). Explain each stage in one or two sentences and name the main source files.

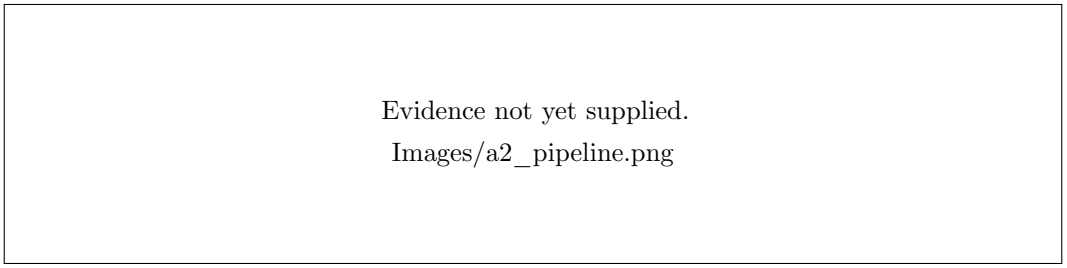


Figure 4: Training and evaluation pipeline for SUIM segmentation.

2.2 Model architectures

2.2.1 U-Net baseline

To complete: Describe the encoder and decoder depth, channel widths, skip connections, normalization, and output head. Give the parameter count. State that the model is implemented by the group and trained from scratch. List its strengths and limits for this task.

2.2.2 Pretrained DeepLabV3

To complete: Describe the ResNet backbone, atrous convolution, the ASPP module, the output stride, and the new eight-class head. Name the pretrained weights (torchvision COCO_WITH_VOC_LABELS_V1) and the library. Give the parameter count and state which parts come from torchvision and which were written by the group.

Table 3: Compared models. Implementation states whether the group wrote the model or used a library.

Model	Role	Initialization	Implementation	Parameters
U-Net	Baseline	Random	Group	*Pending*
DeepLabV3	Pretrained model	*Pending*	torchvision	*Pending*

2.3 Fine-tuning procedure

To complete: Explain how the pretrained model is adapted: replaced classifier head, which layers are trained, separate learning rates for the backbone and the head, and why this choice was made.

2.4 Loss and evaluation metrics

To complete: Define pixel-wise cross-entropy and the class-averaged soft Dice loss, and the combined loss $\mathcal{L} = 0.5\,\mathcal{L}_{\text{CE}} + 0.5\,\mathcal{L}_{\text{Dice}}$. Define per-class IoU $\text{IoU}_k = \text{TP}_k/(\text{TP}_k + \text{FP}_k + \text{FN}_k)$ and Dice $= 2\text{TP}_k/(2\text{TP}_k + \text{FP}_k + \text{FN}_k)$, with confusion counts accumulated over the whole split. State how mIoU and mean Dice average over the eight classes and how absent classes are treated.

2.5 Training configuration

To complete: Fill Table 4 from the configs in configs/segmentation/suim/ and the saved launch configs. Explain how hyperparameters were chosen. Note any difference between a config file and the reported run, such as batch size.

Table 4: Training settings for the reported runs.

Setting	U-Net	DeepLabV3
Input size	*Pending*	*Pending*
Optimizer	*Pending*	*Pending*
Learning rate	*Pending*	*Pending*
Scheduler	*Pending*	*Pending*
Batch size (effective)	*Pending*	*Pending*
Epochs / seed	*Pending*	*Pending*
Weight decay	*Pending*	*Pending*
Loss	*Pending*	*Pending*
Mixed precision	*Pending*	*Pending*
Checkpoint rule	*Pending*	*Pending*
Hardware	*Pending*	*Pending*

2.6 Experimental design

To complete: For each experiment, state the hypothesis, the changed factor, the fixed factors, the metrics, and the decision criterion (Handbook Section 21). The comparisons are U-Net against DeepLabV3, and the controlled experiment of cross-entropy against cross-entropy plus Dice on DeepLabV3. Define in advance what counts as a meaningful difference given a single seed.

Table 5: Experimental design. Each row changes one factor and fixes the rest.

Experiment	Hypothesis	Changed factor	Fixed factors	Decision criterion
Baseline vs. pretrained	*Pending*	U-Net vs. DeepLabV3	*Pending*	*Pending*
Loss function	*Pending*	CE vs. CE + Dice	*Pending*	*Pending*

3 Implementation results

3.1 Training behavior

To complete: Describe the loss and validation mIoU curves for each run: the selected epoch, signs of overfitting or underfitting, and training time. Record implementation issues found during training and how they were fixed.

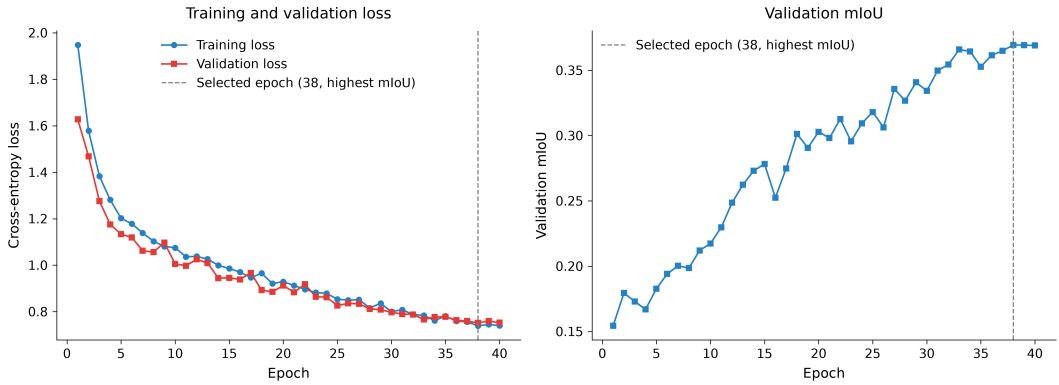


Figure 5: U-Net training loss and validation mIoU on SUIM, seed 42.

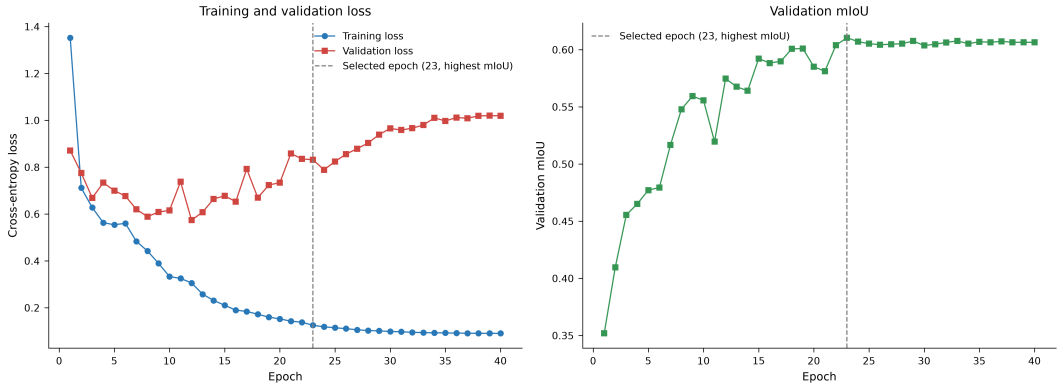


Figure 6: DeepLabV3 with cross-entropy: training loss and validation mIoU on SUIM, seed 42.

Evidence not yet supplied.
Images/deeplabv3_dice_learning_curves.png

Figure 7: DeepLabV3 with cross-entropy plus Dice: training loss and validation mIoU on SUIM, seed 42.

3.2 Quantitative results

To complete: Report test mIoU and mean Dice for every model, using the checkpoint selected on validation. Give per-class IoU, parameter counts, training time, and inference cost. State that each configuration uses one seed, or give mean and standard deviation if more seeds are run.

Table 6: SUIM test results with the checkpoint selected on validation. Each configuration uses seed 42.

Model	Loss	mIoU (%)	Mean Dice (%)	Parameters
U-Net	CE	*Pending*	*Pending*	*Pending*
DeepLabV3	CE	*Pending*	*Pending*	*Pending*
DeepLabV3	CE + Dice	*Pending*	*Pending*	*Pending*

Table 7: Per-class test IoU (%) on SUIM.

Model	BW	HD	PF	WR	RO	RI	FV	SR
U-Net (CE)	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*
DeepLabV3 (CE)	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*
DeepLabV3 (CE + Dice)	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*

3.3 Compute cost

To complete: Compare parameters, training time, inference latency, throughput, and peak memory. Name the hardware for each run (RTX 4060 Laptop or 2× RTX 3090). Explain why time measured on different hardware cannot be compared directly, and remeasure inference on one device if possible.

Table 8: Compute cost. Training time is measured on the hardware listed; inference is measured on *Pending* at batch size *Pending*.

Model	Parameters	Train time (s)	Latency (ms/image)	Peak memory (GB)	Training hardware
U-Net	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*
DeepLabV3 (CE)	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*
DeepLabV3 (CE + Dice)	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*

3.4 Qualitative results

To complete: Show test images with ground truth and predictions from each model: correct predictions, difficult but correct cases, and failures. State the rule used to select the examples so the selection is reproducible.

3.5 Controlled experiment: loss function

To complete: Report the cross-entropy against cross-entropy plus Dice result on DeepLabV3. Compare mIoU, mean Dice, and the IoU of the rare classes RO, PF and WR. State whether the hypothesis is supported under the decision criterion in Section 2.6.

3.6 Error analysis

To complete: Group the errors, for example PF predicted as RI, boundary errors, small objects missed, and background confusion. For each group give a description, its frequency or rate from the confusion matrix, examples, a hypothesized cause, and an improvement idea.

3.7 Comparison and discussion

Evidence not yet supplied.
Images/a2_test_examples.png

Figure 8: Test images, ground truth, and predictions from U-Net and DeepLabV3. State the selection rule.

Table 9: Controlled loss experiment on DeepLabV3. Only the loss changes; data, split, schedule, and seed are fixed. Values are test-set percentages.

Loss	mIoU	Mean Dice	RO IoU	PF IoU	WR IoU
CE	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*
CE + Dice	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*
Difference	*Pending*	*Pending*	*Pending*	*Pending*	*Pending*

Evidence not yet supplied.
Images/deeplabv3_test_confusion_matrix.png

Figure 9: Row-normalized confusion matrix of the best model on the SUIM test set.

To complete: Explain which model is better and by how much, where the differences appear (which classes and region sizes), and likely causes in data, architecture, loss, or training. Discuss accuracy, speed, and parameter trade-offs. Say which conclusions cannot yet be drawn. Listing the highest score is not enough.

3.8 Limitations and conclusion

To complete: List data, method, and experimental limits (single seed, different hardware, source overlap, resizing). Discuss generalization and future work. End with a short answer to the research question.

AI usage disclosure

To complete: Summarize how the group used AI tools in Assignment 2: tool and model, member, stage, purpose, affected files, and what the group checked. Keep this consistent with the A2 section of AI_USAGE.md and the assignment page. Do not use AI to generate data, results, or references.

References

[1] Le Thanh Sach, *Course Project Handbook: Assignments 1–3, Deep Learning and Its Applications, CO3133, Semester 261*, HCMUT.

[2] Group Placeholder, *CO3133 Deep Learning course project repository*. https://github.com/vietanh1802/CO3133-Deep-Learning-261-Placeholder.

[3] Md Jahidul Islam, Chelsey Edge, Yuyang Xiao, Peigen Luo, Muntaqim Mehtaz, Christopher Morse, Sadman Sakib Enan, and Junaed Sattar, *Semantic Segmentation of Underwater Imagery: Dataset and Benchmark*, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020. https://arxiv.org/abs/2004.01241.

[4] IRVLab, University of Minnesota, *SUIM dataset*. https://irvlab.cs.umn.edu/resources/suim-dataset.

To complete: Add the access date.

[5] Olaf Ronneberger, Philipp Fischer, and Thomas Brox, *U-Net: Convolutional Networks for Biomedical Image Segmentation*, MICCAI, 2015. https://arxiv.org/abs/1505.04597.

[6] Liang-Chieh Chen, George Papandreou, Florian Schroff, and Hartwig Adam, *Rethinking Atrous Convolution for Semantic Image Segmentation*, arXiv:1706.05587, 2017. https://arxiv.org/abs/1706.05587.

[7] PyTorch contributors, *torchvision segmentation models: DeepLabV3*. https://docs.pytorch.org/vision/stable/models/deeplabv3.html.

To complete: Add the access date and torchvision version.